

Counting solutions modulo p

A polynomial $f(x_1, \dots, x_n)$ with integer coefficients is said to be *absolutely irreducible* if it cannot be factored into factors of smaller degree in *any* extension field of the rational numbers. For example:

- A polynomial in 1 variable is absolutely irreducible if and only if it has degree 1.
- A homogeneous polynomial in 2 variables is absolutely irreducible if and only if it has degree 1.
- The polynomial $x^2 + y$ in two variables x and y is absolutely irreducible.
- The polynomial $x^3 + y^3 + z^3$ is absolutely irreducible.

The last two statements can be checked using Gauss' lemma.

A fundamental theorem (proved by Nisnevich and Lang-Weil) is:

Theorem: Given f is absolutely irreducible, there is a bound N_0 depending on f such that for any prime p larger than N_0 , there is a solution of $f = 0$ modulo p .

Note that, if $F(x_0, \dots, x_n)$ is homogeneous and absolutely irreducible and $F(1, 0, \dots, 0)$ is non-zero, then $f(x_1, \dots, x_n) = F(1, x_1, \dots, x_n)$ is also absolutely irreducible. Thus, the above theorem also implies the following.

Theorem: Given F is homogeneous and absolutely irreducible, there is a bound N_1 depending on F such that for any prime p larger than N_1 , there is a non-trivial solution of F modulo p .

Recall, that a non-trivial solution of a homogeneous polynomial is one where at least one variable is set to a non-zero value.

Let $N(f, p)$ denote the number of solution of f with entries in $\mathbb{F}_p$. When f is linear, it is clear that $N(f, p) = p^{n-1}$ when there is a coefficient of some variable in f which is not divisible by p .

Theorem: Given f is absolutely irreducible, there is a number $C(f)$ depending only on f such that

$$|N(f, p) - p^{n-1}| \leq C(f)p^{n-1-(1/2)}$$

This stronger version of the previous theorem is how the former is proved.

An identity for $N(f, p)$

Let ζ be a primitive p -th root of unity, so that it is a root of the polynomial $\Phi_p(X) = 1 + X + \dots + X^{p-1}$. It follows that $\sum_{a=0}^{p-1} \zeta^a = 0$. Given any integer c , we see that

$$\sum_{a=0}^{p-1} \zeta^{ac} = \begin{cases} p & \text{if } c \bmod p = 0 \\ 0 & \text{gcd}(c, p) = 1 \end{cases}$$

We now apply this for $f(x_1, \dots, x_n) = c$ to get the identity

$$pN(f, p) = \sum_{(x_1, \dots, x_n) \in \mathbb{F}_p^n} \sum_{a=0}^{p-1} \zeta^{af(x_1, \dots, x_n)} = \sum_{a=0}^{p-1} \sum_{(x_1, \dots, x_n) \in \mathbb{F}_p^n} \zeta^{af(x_1, \dots, x_n)}$$

Now, when $a = 0$, we have $\sum_{(x_1, \dots, x_n) \in \mathbb{F}_p^n} \zeta^{0f(x_1, \dots, x_n)} = p^n$. Thus, we can write this as

$$N(f, p) = p^{n-1} + \frac{1}{p} \sum_{a=1}^{p-1} \sum_{(x_1, \dots, x_n) \in \mathbb{F}_p^n} \zeta^{af(x_1, \dots, x_n)}$$

Thus, the above theorems can be proved by making the following estimate of the latter sum.

Theorem: Given f is absolutely irreducible, there is a number $C(f)$ depending only on f such that

$$\left| \sum_{a=1}^{p-1} \sum_{(x_1, \dots, x_n) \in \mathbb{F}_p^n} \zeta^{af(x_1, \dots, x_n)} \right| \leq C(f)p^{n-(1/2)}$$

The usual proof of this theorem for a general f requires some deeper study of algebraic geometry than available to us in this course. Thus, we will limit our calculations to the special case

$$f(x_1, \dots, x_n) = a_1 x_1^{d_1} + \dots + a_n x_n^{d_n}$$

If $a_i = 0$ for some i then *all* values of x_i will give the same number of roots. A factor of p can this be removed from both sides. Thus, we may assume that $a_1 \dots a_n \neq 0$.

Reduction to counting roots

Since we only want the result for large primes p , we can restrict our attention to primes that do not divide the product $a_1 \dots a_n$. We calculate that

$$\sum_{a=1}^{p-1} \sum_{(x_1, \dots, x_n) \in \mathbb{F}_p^n} \zeta^{af(x_1, \dots, x_n)} = \sum_{a=1}^{p-1} \sum_{(x_1, \dots, x_n) \in \mathbb{F}_p^n} \prod_{i=1}^n \zeta^{aa_i x_i^{d_i}}$$

By induction on n we see that

$$\sum_{(x_1, \dots, x_n) \in \mathbb{F}_p^n} \prod_{i=1}^n \zeta^{aa_i x_i^{d_i}} = \prod_{i=1}^n \sum_{x \in \mathbb{F}_p} \zeta^{aa_i x^{d_i}}$$

where $aa_i \bmod p \neq 0$ for all i .

To see how this works, we perform the first step below:

$$\begin{aligned} \sum_{(x_1, \dots, x_n) \in \mathbb{F}_p^n} \prod_{i=1}^n \zeta^{aa_i x_i^{d_i}} &= \sum_{(x_1, \dots, x_{n-1}) \in \mathbb{F}_p^{n-1}} \sum_{x \in \mathbb{F}_p} \zeta^{aa_n x^{d_n}} \left(\prod_{i=1}^{n-1} \zeta^{aa_i x_i^{d_i}} \right) \\ &= \left(\sum_{x \in \mathbb{F}_p} \zeta^{aa_n x^{d_n}} \right) \left(\sum_{(x_1, \dots, x_{n-1}) \in \mathbb{F}_p^{n-1}} \prod_{i=1}^{n-1} \zeta^{aa_i x_i^{d_i}} \right) \end{aligned}$$

Thus we need to study the sums $\sum_{x=0}^{p-1} \zeta^{bx^d}$ for various values of d , where $b \bmod p \neq 0$.

We have

$$\sum_{x \in \mathbb{F}_p} \zeta^{bx^d} = \sum_{y \in \mathbb{F}_p} m(y) \zeta^{by}$$

where $m(y)$ is the number of solutions of the identity $x^d = y$ in $\mathbb{F}_p$. It is clear that $m(0) = 1$. Moreover, $x \mapsto x^d$ is an endomorphism of the multiplicative group $\mathbb{F}_p^*$.

As seen earlier, $\mathbb{F}_p^*$ is a cyclic group of order $(p-1)$. If $\gcd(d, p-1) = r$ then $x \mapsto x^d$ has image of index r . If y is in this image then $m(y) = r$, else $m(y) = 0$.

We now show a way to write the above sum using Dirichlet characters with modulus p .

Counting r -th roots modulo p

Recall that $\mathbb{F}_p^*$ is a cyclic group of order $(p-1)$. Hence, given r dividing $p-1$, an element $y \in \mathbb{F}_p^*$ is an r -th power if and only if $y^{(p-1)/r} = 1$. In other words, y has order dividing $(p-1)/r$.

Given a Dirichlet character χ modulo p which has order r , we note that $\chi(a) = 1$ if only if $(a \bmod p)$ has order dividing $(p-1)/r$. However, this is *not* a characteristic function for this set of integers since $\chi(a)$ is an r -th root of unity for *all* a such that $\gcd(a, p) = 1$.

To resolve this, we again use the method of adding up powers. For $s = 0, \dots, r-1$ we let $\chi_s(a) = \chi(a)^s$. As seen earlier, if $\chi(a)$ is non-trivial (not 1!) root of unity, then $\sum_{s=0}^{r-1} \chi_s(a) = 0$. In fact

$$\sum_{s=0}^{r-1} \chi_s(a) = \begin{cases} r & \text{if } \chi(a) = 1 \\ \frac{1-\chi(a)^r}{1-\chi(a)} = 0 & \text{if } \chi(a) \neq 1 \end{cases}$$

By the above discussion, we see that $m(y) = \sum_{s=0}^{r-1} \chi_s(y)$ for $y \neq 0$. For $y = 0$, this sum is 0, whereas $m(0) = 1$.

Hence, we get

$$\sum_{y \in \mathbb{F}_p} m(y) \zeta^{by} = 1 + \sum_{y \in \mathbb{F}_p^*} \sum_{s=0}^{r-1} \chi_s(y) \zeta^{by}$$

We now re-group the latter sum as follows.

$$\begin{aligned} \sum_{y \in \mathbb{F}_p} m(y) \zeta^{by} &= 1 + \sum_{y \in \mathbb{F}_p^*} \sum_{s=0}^{r-1} \chi_s(y) \zeta^{by} \\ &= 1 + \sum_{y \in \mathbb{F}_p^*} \chi_0(y) \zeta^{by} + \sum_{s=1}^{r-1} \sum_{y \in \mathbb{F}_p^*} \chi_s(y) \zeta^{by} \\ &= \sum_{s=1}^{r-1} \sum_{y \in \mathbb{F}_p^*} \chi_s(y) \zeta^{by} \end{aligned}$$

since $\chi_0(y) = 1$ and $(b \bmod p) \neq 0$ implies that $1 + \sum_{y \in \mathbb{F}_p^*} \zeta^{by} = 0$. Thus,

$$\sum_{y \in \mathbb{F}_p} m(y) \zeta^{by} = \sum_{s=1}^{r-1} \sum_{y \in \mathbb{F}_p^*} \chi_s(y) \zeta^{by} = \sum_{s=1}^{r-1} \sum_{y \in \mathbb{F}_p} \chi_s(y) \zeta^{by}$$

since $\chi_s(0) = 0$.

Primitive Dirichlet characters modulo p of order dividing r

Suppose g is a generator of the cyclic group $\mathbb{F}_p^*$. Since χ has order r , this means that $\chi(g)$ has order r as an element of $\mathbb{C}^*$; in other words, it is a primitive r -th root of unity.

If ρ is a Dirichlet character with modulus p and order dividing r , then $\rho(g) = \chi(g)^s$ for some s . s means that $\rho = \chi_s$ for some s . Moreover, if ρ is *not* the principal character, then $s \in \{1, \dots, r-1\}$.

Secondly, if ρ is a Dirichlet character with modulus p , then its conductor is either 1 or p . This means that if ρ is not the principal character, then it is *automatically* a primitive character.

In other words, χ_s for $s \in \{1, \dots, r-1\}$ give *all* the primitive Dirichlet characters modulo p of order dividing r .

Expression in terms of Gauss sums

Recall that for a Dirichlet character ρ with modulus p , we have the Gauss sum

$$G(k, \rho) = \sum_{a=0}^{p-1} \rho(a) \zeta^{ka}$$

We now combine the calculations of the previous sections as follows.

Given a prime p , a primitive p -th root of unity ζ , an integer b such that $b \bmod p \neq 0$ and an integer d with $r = \gcd(d, p-1)$.

$$\sum_{x \in \mathbb{F}_p} \zeta^{bx^d} = \sum_{s=1}^{r-1} G(b, \chi_s)$$

We can express the latter expression more conceptually as follows.

The sum of $G(b, \rho)$ as ρ ranges over primitive Dirichlet characters of modulus p and order dividing r .

Completing the estimation

We saw that

$$N(f, p) - p^{n-1} = \frac{1}{p} \sum_{a=1}^{p-1} \prod_{i=1}^n \left(\sum_{x \in \mathbb{F}_p} \zeta^{aa_i x^{d_i}} \right)$$

Combining this with the above using $r_i = \gcd(d_i, p-1)$, we have

$$N(f, p) - p^{n-1} = \frac{1}{p} \sum_{a=1}^{p-1} \prod_{i=1}^n \left(\sum_{s=1}^{r_i-1} G(aa_i, \chi_s) \right)$$

We have seen that for primitive Dirichlet characters ρ with modulus m we have $|G(k, \rho)| = m^{1/2}$ for all k such that $\gcd(k, m) = 1$. Taking $m = p$, we apply this to the primitive Dirichlet characters χ_s of modulus p and to $k = aa_i$ which satisfies $\gcd(aa_i, p) = 1$. This gives us the inequality

$$|N(f, p) - p^{n-1}| \leq \frac{1}{p} \sum_{a=1}^{p-1} \prod_{i=1}^n \left((r_i - 1) p^{1/2} \right)$$

Note that $r_i \leq d_i$, so we get

$$|N(f, p) - p^{n-1}| \leq \frac{1}{p} (p-1) p^{n/2} \prod_{i=1}^n (d_i - 1)$$

which proves a stronger bound for $N(f, p)$ than the one stated in the theorem for f which is of the form $a_1 x^{d_1} + \dots + a_n x^{d_n}$ where a_i are coprime to p .